

Pediatric Constipation Management Protocol

Introduction:

Constipation poses a widespread concern in childhood on a global scale, exerting a substantial impact on the social, physical, and emotional well-being of affected children and their caregivers. Functional constipation, constituting 90-95% of all causes, is particularly prevalent, with the peak incidence occurring between the ages of 2 and 4 years. While constipation remains a prevalent issue, the management of many patients often falls short of optimal standards due to a lack of experience in its treatment.

Objective:

To support general practitioners and pediatric healthcare providers in effectively managing children experiencing functional constipation.

Scope:

- General Practitioners
- Pediatric Emergency Physicians
- Pediatricians

Definitions:

- 1- Functional Constipation:
 - o The most accurate definition of functional constipation is based on the Rome IV criteria.

Table 1 Rome IV criteria for functional constipation in children [2, 124]

Age	<4 years	Developmental age of ≥4 years
Rome IV criteria	1. <3 defecations per week 2. History of excessive stool retention 3. History of painful or hard bowel movements 4. History of large-diameter stools 5. Presence of a large fecal mass in the rectum 6. In toilet-trained children, the following additional criteria may be used: ≥1 episode of fecal incontinence per week after the acquisition of toileting skills 7. History of large-diameter stools that may obstruct the toilet Must fulfill ≥2 criteria for ≥1 month prior to diagnosis	1. <3 defecations in the toilet per week 2. ≥1 episode of fecal incontinence per week 3. History of retentive posturing or excessive volitional stool retention 4. History of painful or hard bowel movements 5. Presence of a large fecal mass in the rectum 6. History of large-diameter stools that may obstruct the toilet Must fulfill ≥2 criteria at least once per week for ≥1 month prior to diagnosis Insufficient criteria for diagnosis of irritable bowel syndrome with constipation

*de Geus et al. Pediatric Drugs, 2023

- 2- Intractable Constipation:
 - o Constipation not responding to optimal conventional treatment for at least 3 months

Evaluation:

Evaluation of childhood constipation primarily consists of a thorough medical history and a complete physical examination. Investigation might be considered for patients with red flags (see table 1) for organic causes and patients with refractory constipation.

- 1- History:
 - o Age of onset
 - o Passage of first meconium
 - o Stools: consistency, frequency, and soiling (define it by Bristol Chart)
 - o Ask about alarming symptoms *see table 1

- Past medical history including previous treatment for constipation and compliance and any comorbid (neuromuscular and autoimmune)
 - Family history
 - Nutritional history
 - Social history, including toilet training, stressors
- 2- Examination:
- Growth parameters
 - Abdominal examination
 - Perianal and rectal examinations (anal position, anal fissure, anal stenosis, hard stools in the rectum)
 - Back examination looking for spina bifida
 - Neuromuscular examination
 -
- 3- Investigations:
- If there are no alarming signs; NO investigations are required at initial presentation.
 - Abdominal X-rays ARE NOT INDICATED to make a diagnosis of constipation or to assess the adequacy of cleanout

Treatment:

- 1- Non-Pharmacological (Education and Lifestyle/Dietary modifications):
- It is important to explain to the parents that constipation requires both behavioral changes and medical treatment.
 - Parents should know that treatment should be at least 3 months and compliance is an important factor in ensuring successful treatment
 - Behavioral changes including regular toileting especially after main meals (for 3-5 minutes)
 - Ensure good fiber and fluid intake
- 2- Pharmacological:
- Disimpaction:
 - The first step in pharmacological treatment when there is a hard-fecal mass in the rectum
 - Polyethylene glycol (PEG- movicol available in Oman) should be used as first-line for disimpaction (see table for dosing)
 - If PEG is not available then phosphate/Saline enema- once daily for 3-6 days
 - Inform the child and parents that the treatment can cause abdominal pain and discomfort.
 - Maintenance Therapy
 - Should be started as soon as the bowel is disimpacted.
 - Parents and child should clearly counsel that the treatment should continue for at least 3 months or 1 month after resolution of the symptoms.
 - Main available medications: Lactulose, movicol, Milk of magnesia and Bisacodyl (see table for doses)

Consultations:

To Surgical Team:

- Symptoms or signs suggestive of HD, anorectal malformation or intestinal obstruction
- For manual disimpaction if medically failed or fecal masses causing mechanical obstruction

Referral to Gastroenterology team:

- Patients who has red flags
- Chronic constipation resistant to the standard medical therapy.
 - Make sure that the patient was compliant for medical therapy

- Include the following in your referral:
 - Onset of constipation
 - Patient's growth
 - Medical therapy duration, doses and compliance
 - Alarming symptoms and signs
- Consider the following evaluation while waiting for appointment:
 - Celiac screen (Anti-TTG)
 - Thyroid function test (Free T4 and TSH)
 - Bone profile

Table 1: Red flags suggestive of organic causes of constipation:

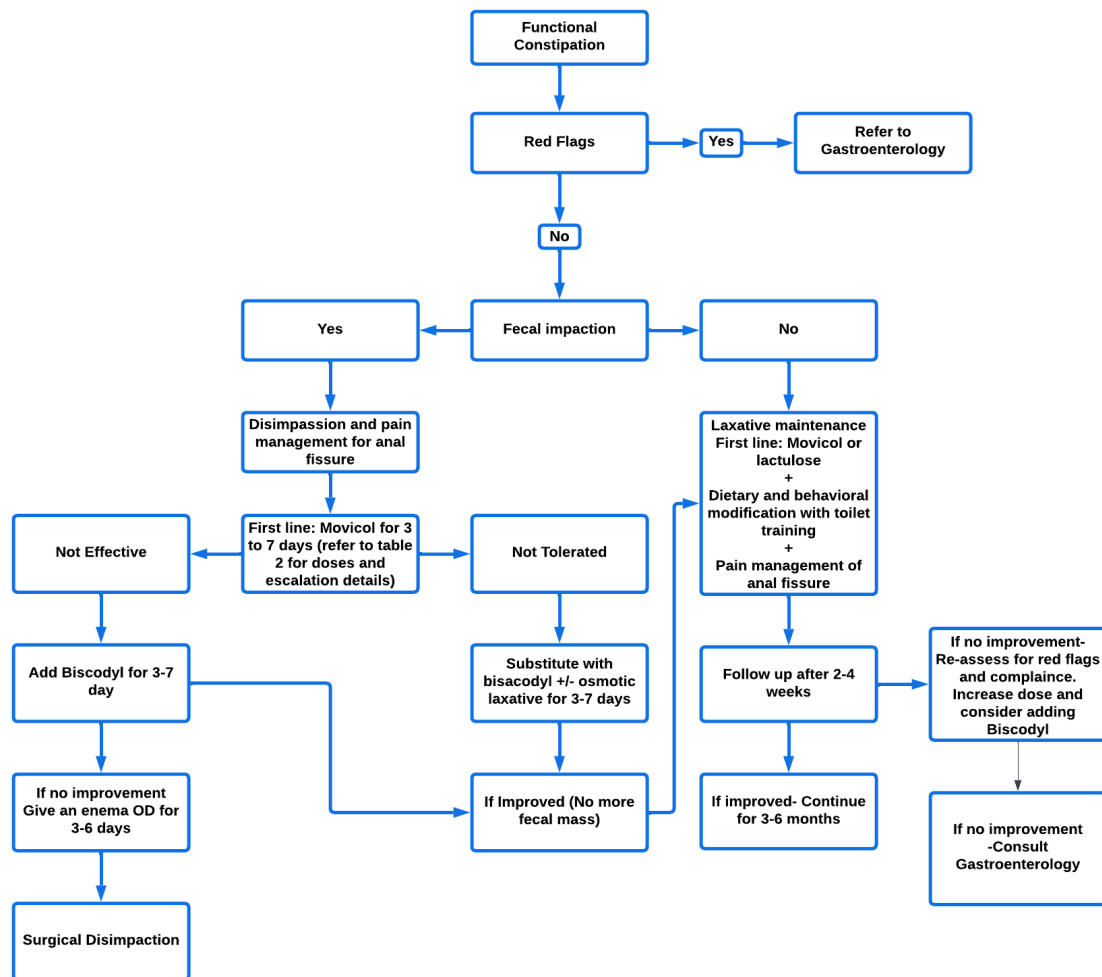
Red Flags	Possible diagnosis
Onset before 1 month	Congenital malformation of anorectum or spine, Hirschsprung disease (HD), hypothyroidism
Delayed passage of meconium (more than 48 hours after birth)	HD, cystic fibrosis (CF), congenital malformation of anorectum or spine
Failure to thrive (look for other systemic causes)	HD, CF, disorders of malabsorption
Abdominal distension	HD
Intermittent diarrhea and explosive stools	
Empty rectum	
Gushing of stool with rectal examination	
Tight anal sphincter	HD, anorectal malformations
Pilonidal dimple covered by tuft of hair	Spinal Cord Abnormality
Midline pigmentary abnormalities of lower spine	
Abnormal neurologic examination (absent anal wink, absent cremasteric reflex, decreased lower extremity reflexes and/or tone)	

➤ Nurko et al. AAFP 2018

Table 2: Medication guide:

Disimpaction Therapy										
	Drug	Dose							Side Effect	
Oral	Movicol (13.8g/sachet)	Increase the dose gradually until complete disimpaction or for 7 days.							Allergic reaction, flatulence	
		Age (years)	D1	D2	D3	D4	D5	D6		D7
		1-5	1	2	2	3	3	4		4
		5-12	2	3	4	5	6	6		6
	Lactulose (if PEG is not available)	4-6 ml/kg/day								
Rectal	Phosphate Enema	1-12 years: 2.5 ml/kg (max 133 ml/dose)							Abdominal and anal pain, dehydration, electrolyte imbalance (HyperNa, Hyper PO4, Hypo Ca, Hypo K)	
	Saline Enema	- < 1 year: 5 ml - > 1 year: 10 ml								
	Bisacodyl Suppository	- > 2years: 5-10 mg								
	Glycerine Suppository	- < 5 years: infant Suppository - > 5 years: adult Suppository								
Maintenance Therapy										
Oral	Movicol	- < 1 year – 0.5 sachet daily - 1-6 years- 1-4 sachets daily (adjust the dose to achieve 1-2 soft stools daily – Max 4 sachet/d								
	Lactulose	- < 1 year: 3-5 ml/day - 1-5 years- 5-10 ml/day - 5-12 years- 10-30 ml/day (1-3 ml/kg/day)							Abdominal distension, discomfort	
	Magnesium Hydroxide (Milk of Magnesia)	- 2–5 y: 0.4–1.2 g/day, once or divided - 6–12 y: 1.2–2.4 g/day, once or divided							Mg overdose (hyper Mg, hyper PO4, hypoCa)	
	Mineral Oil	- 5-15 ml daily							Lipid pneumonia if aspirated.	

Figure 1: Flow chart for management of pediatric constipation:



*Modified from Dr. Nawal Al Rasbi Protocol

Reference:

1. Tabbers MM, DiLorenzo C, Berger MY, Faure C, Langendam MW, Nurko S, et al. Evaluation and treatment of functional constipation in infants and children: evidence-based recommendations from ESPGHAN and NASPGHAN. *J Pediatr Gastroenterol Nutr.* 2014 Feb;58(2):258–74.
2. Constipation in children and young people: diagnosis and management.
3. Sandweiss DR, Allen L, Deneau M, Harnsberger J, Pasmann A, Smout R, et al. Implementing a Standardized Constipation-Management Pathway to Reduce Resource Utilization. *Academic Pediatrics.* 2018 Nov 1;18(8):957–64.
4. Mulhem E, Khondoker F, Kandiah S. Constipation in Children and Adolescents: Evaluation and Treatment. *afp.* 2022 May;105(5):469–78.
5. de Geus A, Koppen IJN, Flint RB, Benninga MA, Tabbers MM. An Update of Pharmacological Management in Children with Functional Constipation. *Paediatr Drugs.* 2023;25(3):343–58.
6. Management protocol of pediatric constipation, Dr. Nawal Al Rasbi, AL- Nahdah Hospital (consent taken from Dr. AL Rasbi taken)

